# Prabhavit Gupta

Forward Deployed Engineer | Autonomous Software Agents, Developer Tools & Enterprise Integrations

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (EU Permanent Resident | Open to Relocation to SF)

## Executive Summary

- **Forward Deployed Engineering & Enterprise Deployments:** Software engineer with 3 years commercial software experience embedding into enterprise engineering environments, architecting production integrations, and deploying autonomous AI agent systems.
- **Autonomous Agents, Developer Tooling & CI/CD:** Deep practical expertise with autonomous tool execution loops, Model Context Protocol (MCP), automated CI/CD workflows, test harnesses (Playwright, PyTest), and diagnosing edge cases across diverse production codebases.
- **High-Agency Execution & Technical Leadership:** Experienced in partnering directly with customer engineering leads, translating ambiguous system requirements into production-ready software, and channeling field deployment insights into core platform improvements.

## Technical Skills & Core Competencies

- **Forward Deployed Engineering:** Customer Deployments, System Architecture, Technical Scoping, API Integrations, CI/CD Pipeline Automation, On-Site Technical Delivery, Codebase Diagnostics.
- **Autonomous Agents & Tooling:** Autonomous Agent Loops, Tool Calling, Model Context Protocol (MCP), Sandboxed Tool Execution, Automated Testing Suites, Prompt Engineering, Evaluation Frameworks (DeepEval, Langfuse).
- **Core Software Development:** Python (AsyncIO, Requests, PyTest), TypeScript, Node.js, Next.js, Laravel, REST APIs, GraphQL, SQL (PostgreSQL, SQLite), Docker, Git.
- **Operations & Developer Tooling:** GitHub Actions, Linux CLI, Postman, Distributed Tracing (Langfuse), Agile/Jira, Technical Documentation.

## Professional Experience

**Forward Deployed AI Engineer & Technical Solutions Consultant** • *Independent Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Embedded directly with client engineering teams to scope technical requirements, architect custom integrations, and deploy autonomous AI agent workflows solving developer bottlenecks.
- Built custom integration harnesses connecting enterprise developer workflows, code repositories, and issue trackers to centralized LLM agent execution services.
- Implemented Model Context Protocol (MCP) tool loops and sandboxed execution runtimes, enabling agents to safely run automated code checks, query databases, and dispatch actions.
- Diagnosed and resolved deployment friction across network proxies, authentication, and heterogeneous developer stacks, accelerating customer onboarding and time-to-value.
- Synthesized recurring customer integration patterns into reusable open-source examples, internal tooling, and architectural documentation.

**Lead Developer & Product Engineer** • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Led platform engineering and integration architecture, scaling the application from 2,000 to 4,500+ active users across European tech hubs.
- Re-architected core application off legacy infrastructure onto Next.js and Laravel, ensuring continuous operational availability, data parity, and zero downtime.
- Built automated data ingestion and extraction pipelines aggregating unstructured venue inventory across multiple European platforms with validation barriers.
- Integrated external third-party management APIs (OfficeRnD, Nexudus) for automated booking orchestration via real-time webhooks.
- Deployed automated conversational support agents connected to live booking databases, reducing human support overhead while maintaining high resolution quality.

**Full-Stack Solutions Engineer** • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Delivered custom API integrations and automated backend microservices in Python and TypeScript, connecting client platforms with internal data warehouses.
- Built automated unit and integration testing suites within Docker CI/CD pipelines, establishing reliable integration baselines.

- Collaborated directly with technical and business stakeholders to translate manual operational workflows into reliable software specifications.

## Key Engineering Projects

- **Autonomous Tool-Calling & Evaluation Harness (TypeScript + MCP + Langfuse):** Architected an execution harness enabling LLMs to safely invoke external tools, validating payloads against strict schemas with automated tracing and regression metrics.
- **Universal Multi-Source Ingestion Pipeline (Python + Playwright + REST APIs):** Designed an automated data harvesting and enrichment system parsing 900+ structured and semi-structured external sources with schema validation and error isolation.

## Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  *Rigorous foundation in formal logic, epistemology, decision theory, and complex systems analysis.*
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). Open to international relocation to San Francisco with visa sponsorship; available immediately.